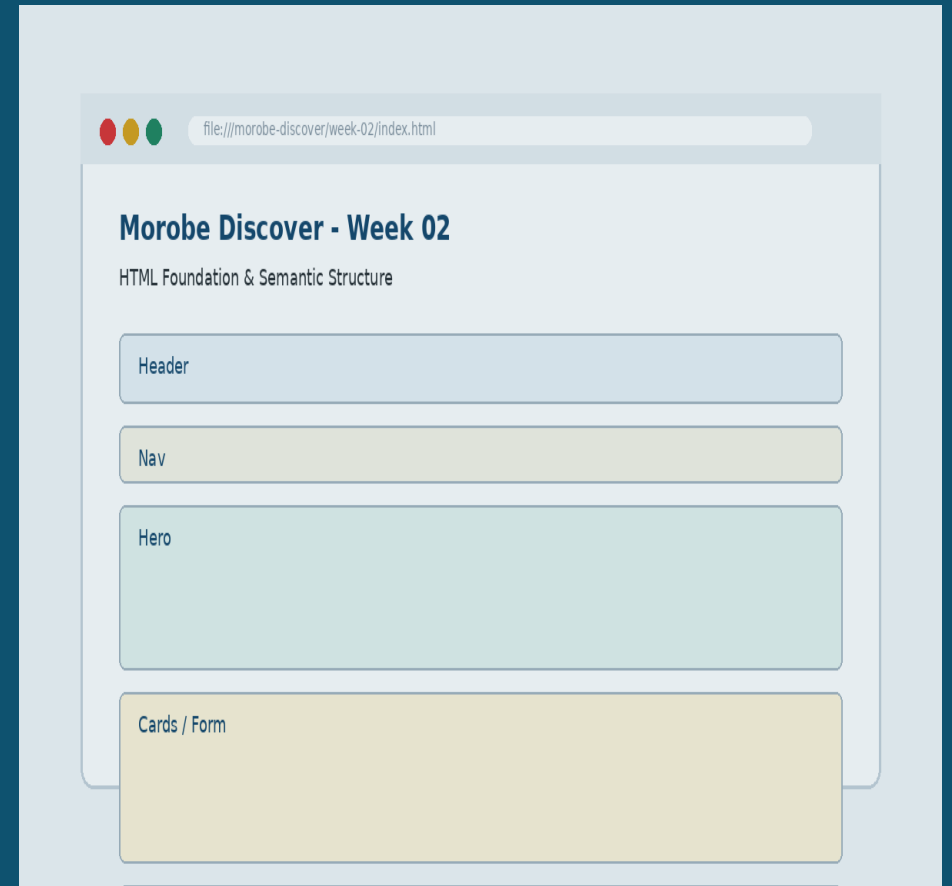


Week 02

HTML Foundation & Semantic Structure

Create meaningful page regions and build web pages using appropriate structural elements.



Learning outcomes

Create meaningful page regions and build web pages using appropriate structural elements.

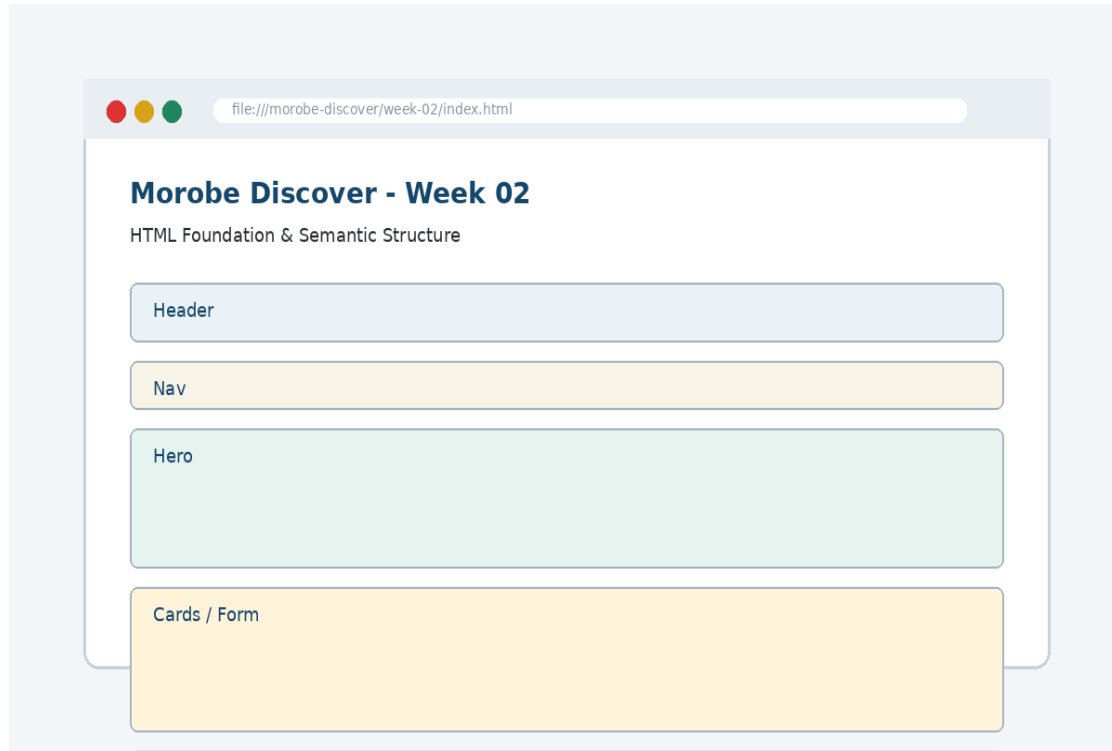
Explain the core programming/design concept for this week

Read and modify key code patterns

Test the weekly project increment

Prepare evidence for GitHub submission

Today's problem



Why it matters

HTML should describe the meaning and structure of the content before CSS changes the appearance.

Project before -> after

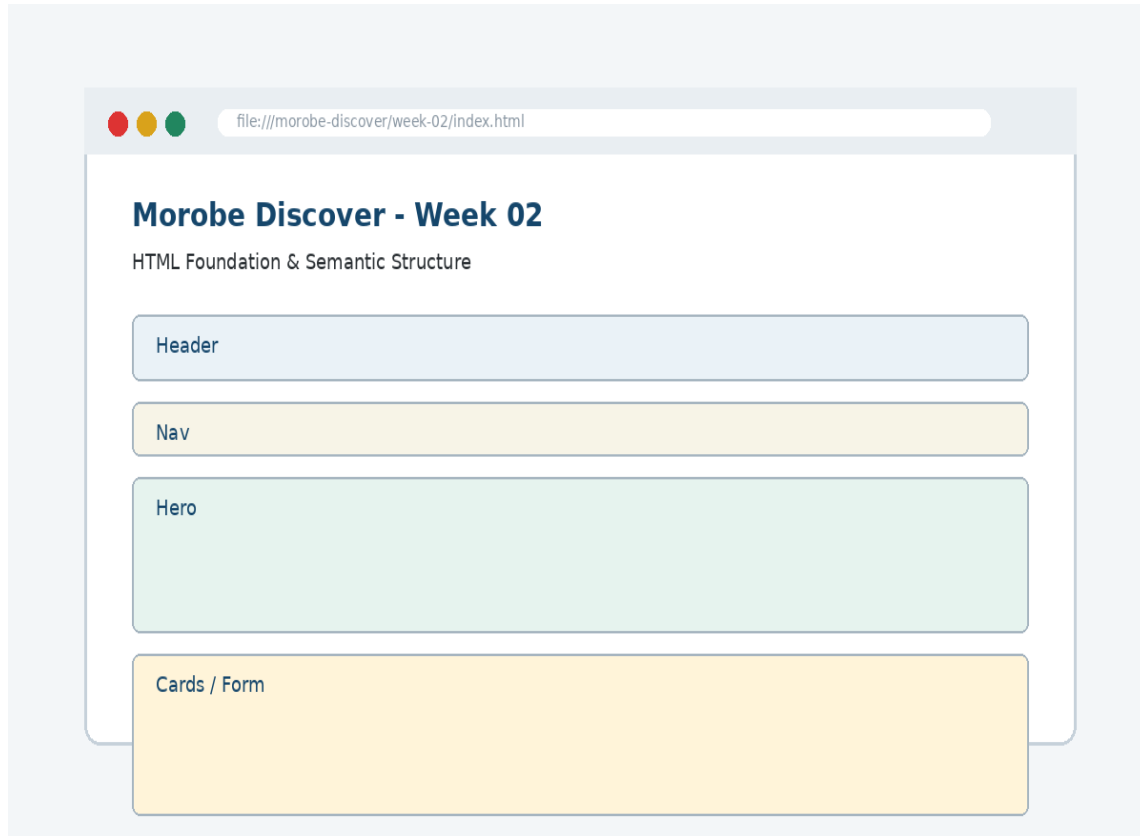
Before Week 02

Week 01 produced the sitemap, wireframe and repository scaffold.

After Week 02

The homepage is coded as a meaningful HTML5 structure that reflects the wireframe.

What we build this week



Create the HTML document shell
Add title and metadata
Translate wireframe regions into semantic elements
Add navigation links and headings
Add placeholder content for hero, cards and footer

Core principle

HTML should describe the meaning and structure of the content before CSS changes the appearance.

Keep in mind

A good web developer can point to the code, the browser result, and the user reason behind the decision.

Key concepts

Concept 1

The document shell gives the page its valid base

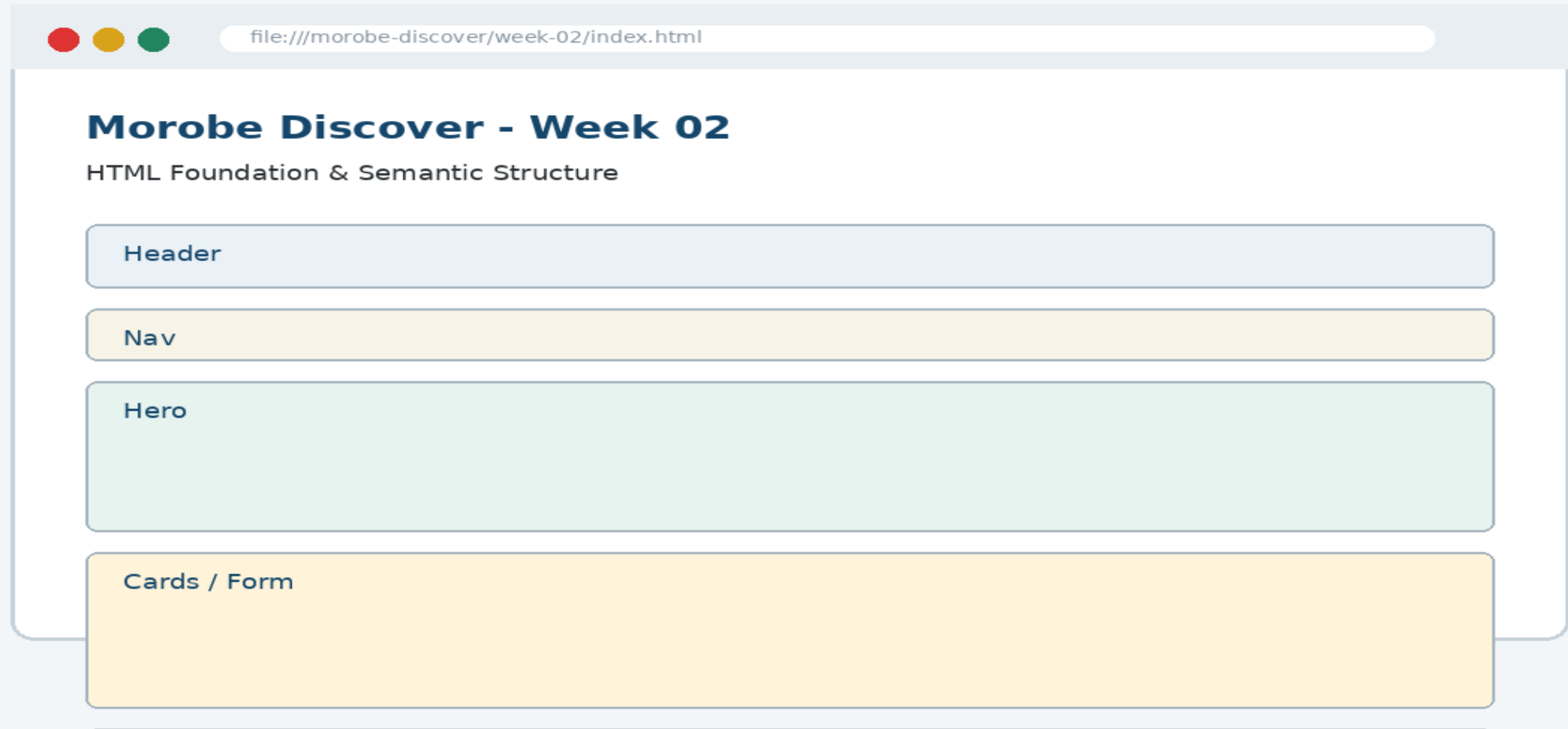
Concept 2

Semantic elements name regions by purpose

Concept 3

Heading order creates a document outline

Project anatomy



Key code pattern A

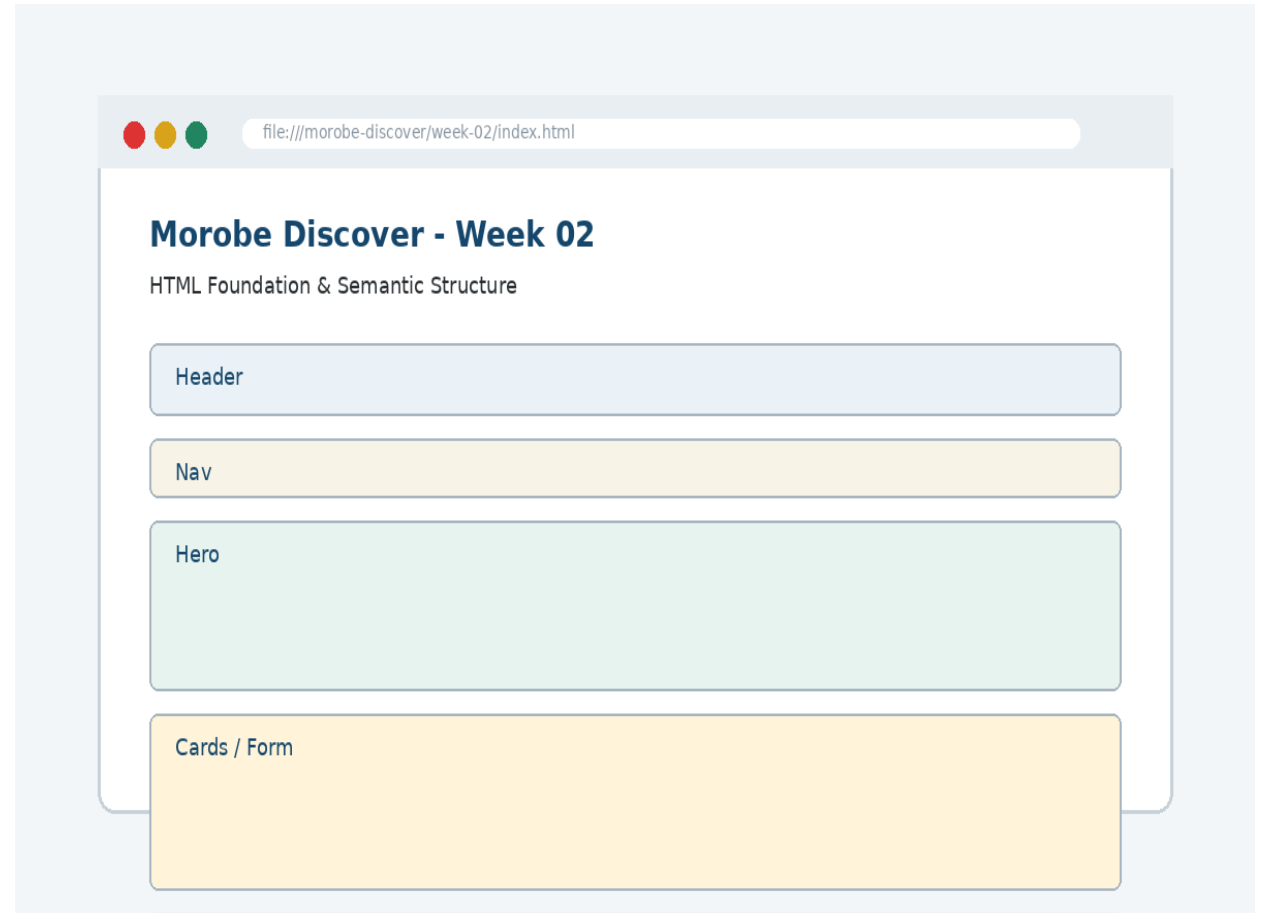
```
<header>
  <h1>Morobe Discover</h1>
</header>
<nav>...</nav>
<main>...</main>
<footer>...</footer>
```

Explain it

Which file contains this code?
Which line creates the visible change?
What should appear in the browser?

Code -> visual result

```
<header>  
  <h1>Morobe Discover</h1>  
</header>  
<nav>...</nav>  
<main>...</main>
```



What changed?

Project file changed for HTML Foundation & Semantic Structure

The browser output is now closer to the required weekly evidence

The change supports a user task or quality requirement

The change can be tested and committed

Key code pattern B

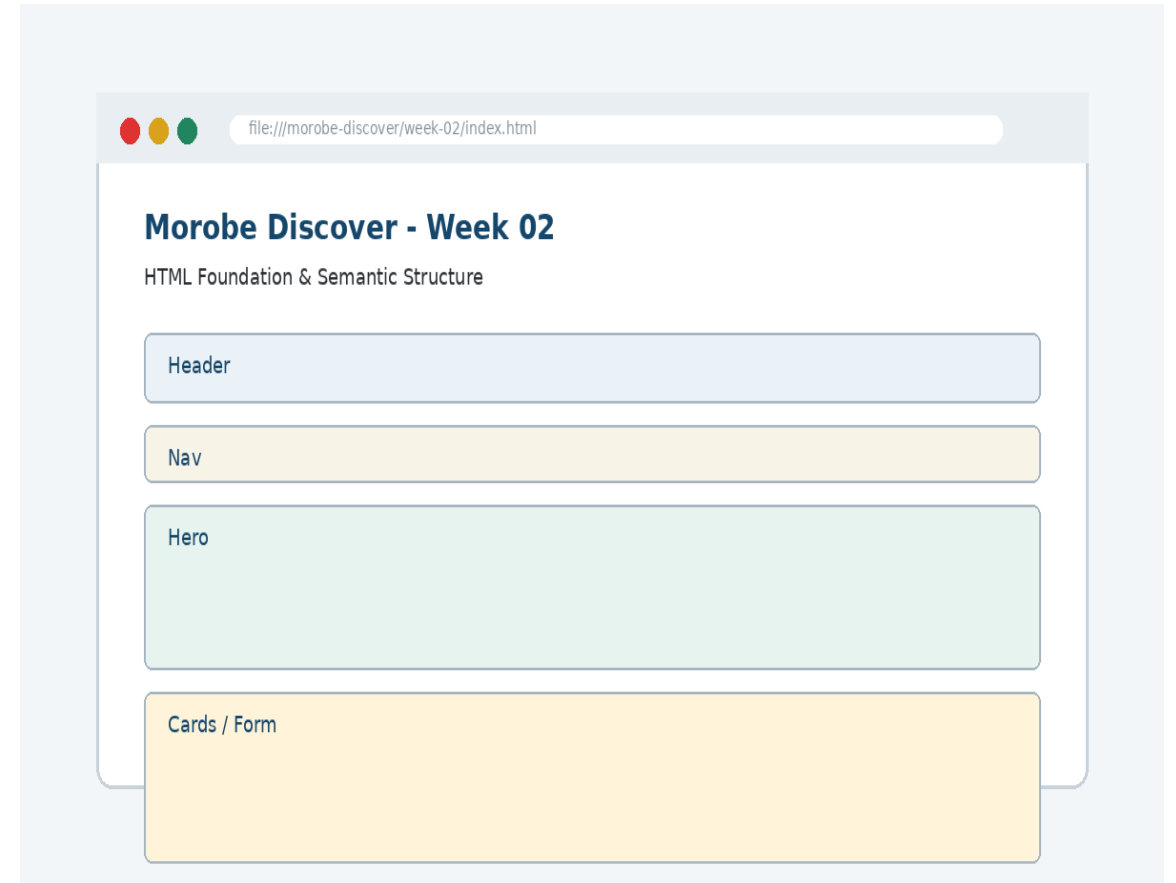
```
<section class="hero">
  <h2>Discover Morobe</h2>
  <p>Find destinations, events and trip
ideas.</p>
</section>
```

Why this pattern is needed

The second pattern usually applies the same idea to a different component, state, layout or data flow.

Apply it to Morobe Discover

Create the HTML document shell
Add title and metadata
Translate wireframe regions into semantic elements
Add navigation links and headings



Compare the approaches

Weak approach

- Memorise the definition
- Copy code without understanding
- No browser evidence
- No test note

Professional approach

- Explain the concept
- Apply the correct code pattern
- Inspect the result
- Commit and describe the evidence

Common mistake

Mistake

The page appears unchanged, but the student assumes the feature is complete.

Correction

Check the correct file, refresh the browser, inspect DevTools, and verify the expected output.

Why does this fail?

```
// Broken checkpoint  
// File changed, but the browser result does not  
change.  
// Possible causes:  
// - wrong file  
// - file not linked  
// - syntax error  
// - browser not refreshed
```

Class question

Which cause would you check first, and why?

Fix the problem

- Confirm the browser is opening the correct file
- Check links to CSS/JS/data files
- Inspect the console for errors
- Use a small working example
- Test again before committing

Class activity

Activity - 7 minutes

Look at the wireframe and label which region becomes header, nav, main, section and footer.

Work in pairs

Point to the project evidence

Prepare one answer to share

Implementation steps 1-3

Create the HTML document shell

Add title and metadata

Translate wireframe regions into semantic elements

Implementation steps 4-5

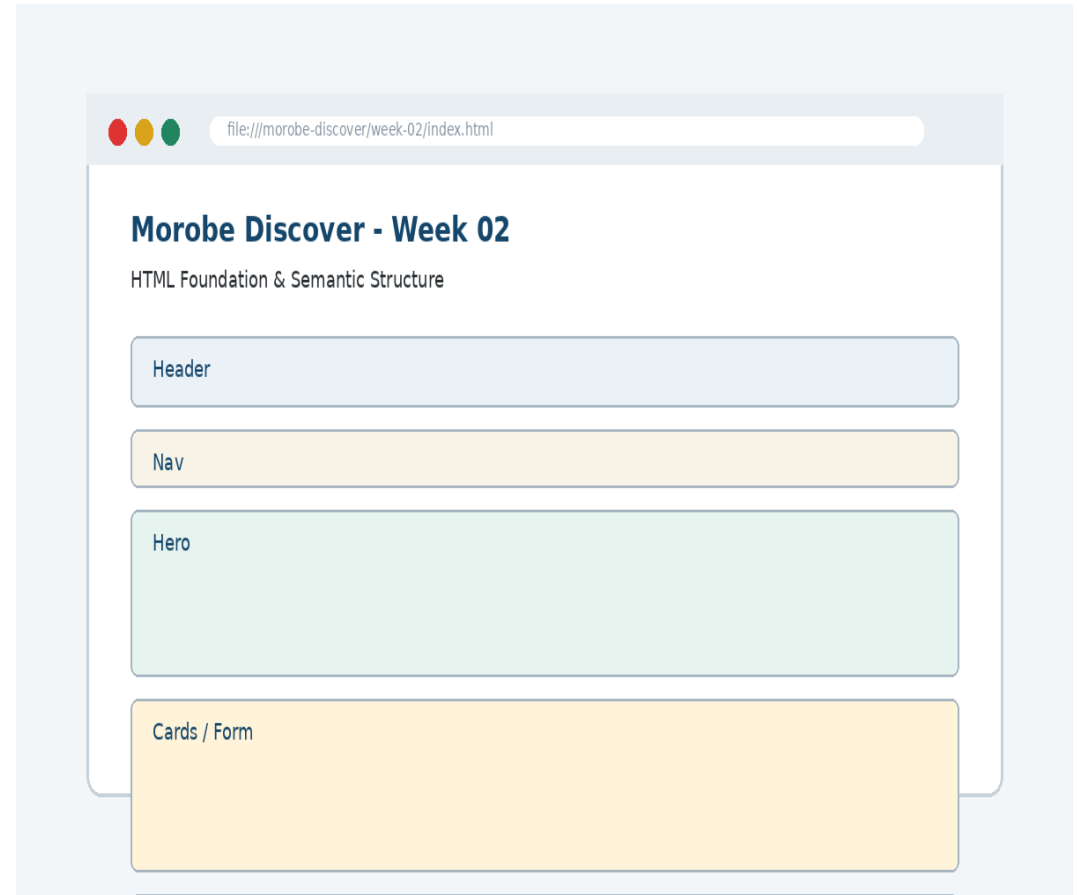
Add navigation links and headings

Add placeholder content for hero, cards and footer

LIVE DEMO

Demo checklist

1. Open the current project
2. Locate the affected file
3. Add or modify the key code
4. Refresh the browser
5. Inspect the result
6. Explain the change



Predict the output

```
<section class="hero">
  <h2>Discover Morobe</h2>
  <p>Find destinations, events and trip
ideas.</p>
</section>
```

Prediction prompt

What should change on screen when this code is applied correctly?

Browser result



Testing the feature

Validate the document structure

Check one h1 per page

Check navigation links point to planned pages

Inspect the browser output

Inspect with browser tools

Inspection path

Right click -> Inspect
Elements panel -> structure/classes
Console -> JavaScript errors
Network/Application -> data/storage when relevant

Evidence to save

Screenshot
Short test note
Changed files
Git commit message

Git evidence

```
git add .  
git commit -m "Week 02 HTML Foundation & Semantic  
Structure project increment"
```

Why it matters

Git history shows progression from Week 01 to final deployment. It is part of the assessment evidence.

Complete source code

Repository:
`https://github.com/YOUR-USERNAME/is229-morobe-discover`

Weekly folder:
`/week-02/`

Complete project:
`/complete-project/`

Reminder

Do not paste a whole file into a slide. Use the repository for complete code.

Mini challenge

Challenge

Modify one part of the Week 02 implementation and explain what changed in the browser.

State the file edited

State the line/section changed

Show the before and after result

What counts as evidence?

Technical evidence

Changed code files
Browser screenshot
Test note
Git commit

Understanding evidence

Code explanation
Why this method fits
Connection to previous week
Connection to next week

Project lab handoff

Create the HTML document shell

Add title and metadata

Translate wireframe regions into semantic elements

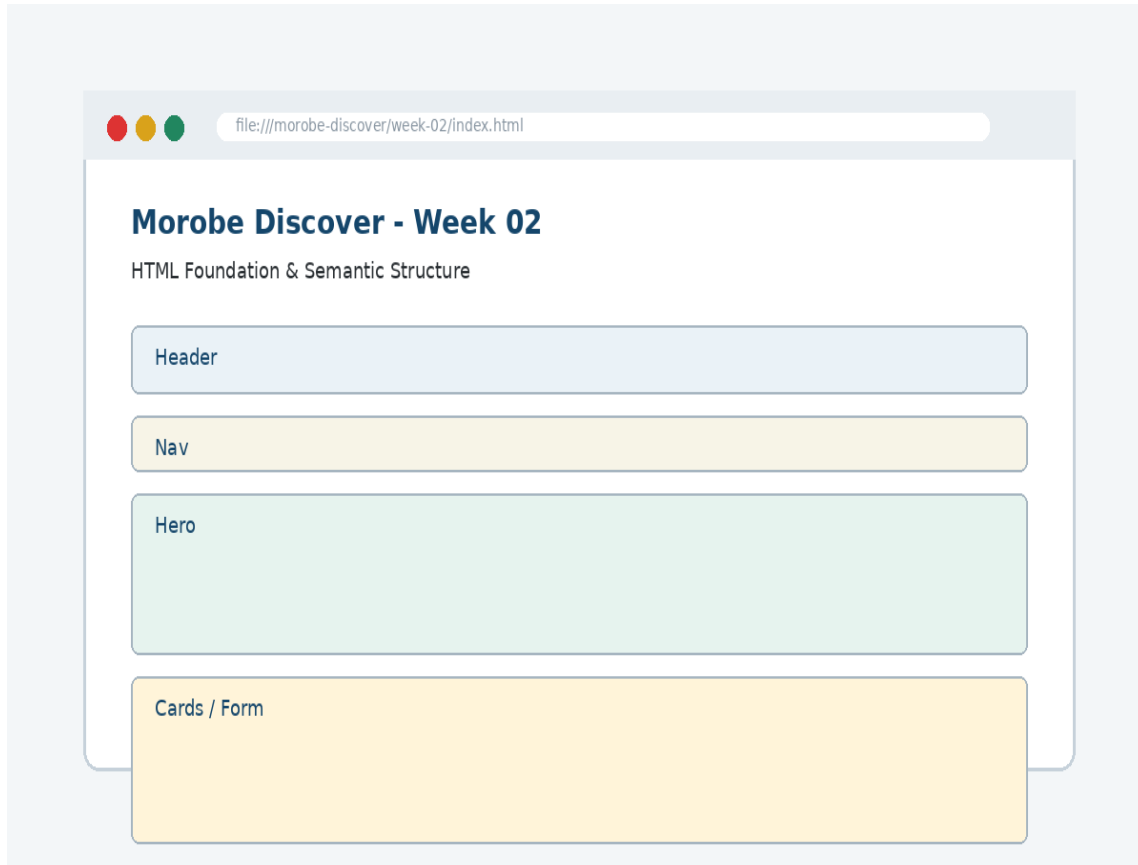
Add navigation links and headings

Add placeholder content for hero, cards and footer

Definition of done

Week 02 feature is implemented
Project still preserves earlier week features
Browser result matches the expected output
Test note and screenshot are saved
Git commit message is meaningful

What we built today



The homepage is coded as a meaningful HTML5 structure that reflects the wireframe.

Evidence: Semantic HTML skeleton using header, nav, main, section, footer and a clear document outline

Technology: HTML

Lecture summary

HTML should describe the meaning and structure of the content before CSS changes the appearance.

The document shell gives the page its valid base

Semantic elements name regions by purpose

Testing and Git are part of development, not extras

Prepare for next class

Finish this week's lab before the next class
Read the next topic in advance
Bring the current project ready to extend
Be ready to explain one code decision

Exit question

How did Week 02 improve Morobe Discover, and what will the next week build on?